Finding the density of the vertical coordinate of a random point on a circle

A detailed worked solution using cumulative distribution functions

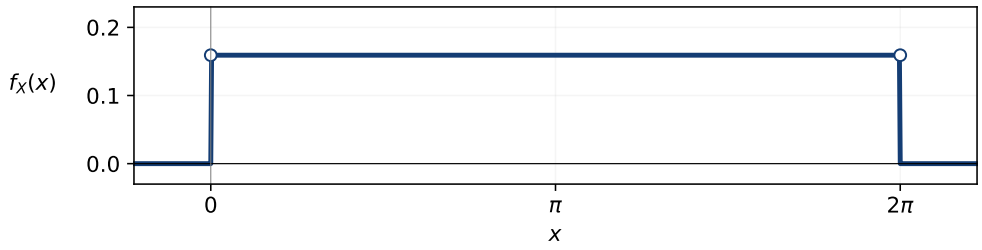
Problem. A point P is selected uniformly at random on the circumference of the unit circle. Let Y denote its vertical coordinate. Find the cumulative distribution function (CDF) and probability density function (PDF) of Y.

Meaning of "uniformly on the circle". All arcs of the same length are equally likely to contain P. Parameterize the point by an angle X, measured counterclockwise from $(1, 0)$, so that $P = (\cos X, \sin X)$. On the *unit* circle an arc subtending an angle of θ radians has length θ. Since the total circumference is 2π, an arc of length θ has probability $\theta/(2\pi)$. Consequently X is uniformly distributed over any interval of length 2π that describes the circle exactly once (apart from duplicated endpoints, which have probability zero).

Solution A. Take $X \sim \text{Uniform}(0, 2\pi)$

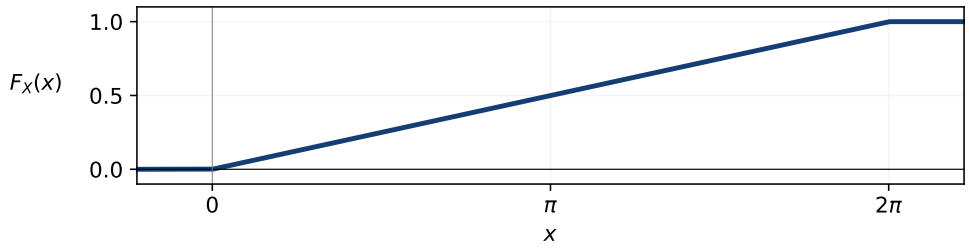
Step 1: Write the PDF of X as a piecewise function.

$$f_X(x) = \begin{cases} \dfrac{1}{2\pi}, & 0 < x < 2\pi, \\ 0, & \text{otherwise.} \end{cases}$$



Step 2: Integrate to obtain the CDF of X. Recall $F_X(x) = \mathbb{P}(X \le x) = \int_{-\infty}^{x} f_X(t)\,dt$. Thus

$$F_X(x) = \begin{cases} 0, & x \le 0, \\ \dfrac{x}{2\pi}, & 0 < x < 2\pi, \\ 1, & x \ge 2\pi. \end{cases}$$



Step 3: Define the new random variable. Since Y is the vertical coordinate of $P = (\cos X, \sin X)$,

$$\boxed{Y = \sin X.}$$

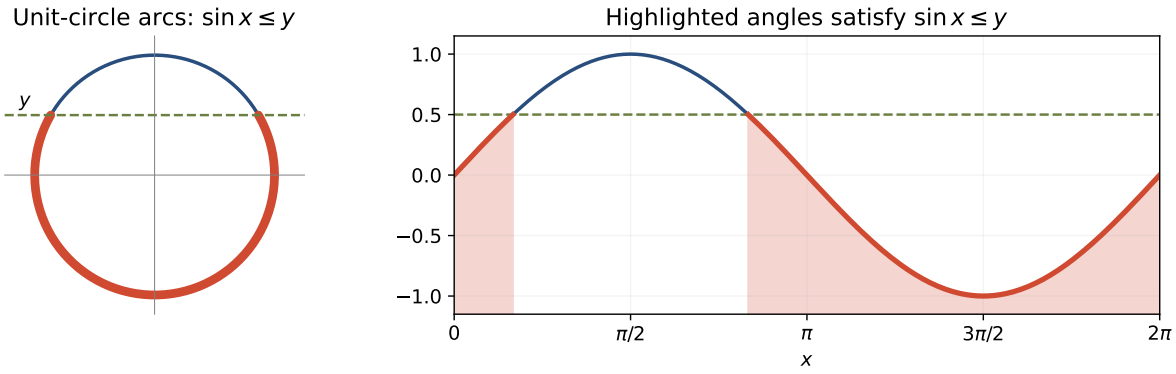
Because $-1 \le \sin X \le 1$, the CDF must have the piecewise form

$$F_Y(y) = \begin{cases} 0, & y \le -1, \\ \underbrace{\mathbb{P}(\sin X \le y)}_{\text{to be evaluated by locating angles}}, & -1 < y < 1, \\ 1, & y \ge 1. \end{cases}$$

Step 4: Compute the middle expression carefully. Put $a = \arcsin y$, where $a \in (-\pi/2, \pi/2)$. We seek the angles x for which $\sin x \le y$. The sine function is not one-to-one over $(0, 2\pi)$; therefore merely writing $x \le \arcsin y$ would be incorrect.

Case A1: $0 < y < 1$, so $0 < a < \pi/2$. The equality $\sin x = y$ occurs at $x = a$ and $x = \pi - a$ in $(0, 2\pi)$. The angles satisfying the inequality are

$$x \in (0, a] \ \cup \ [\pi - a, 2\pi).$$

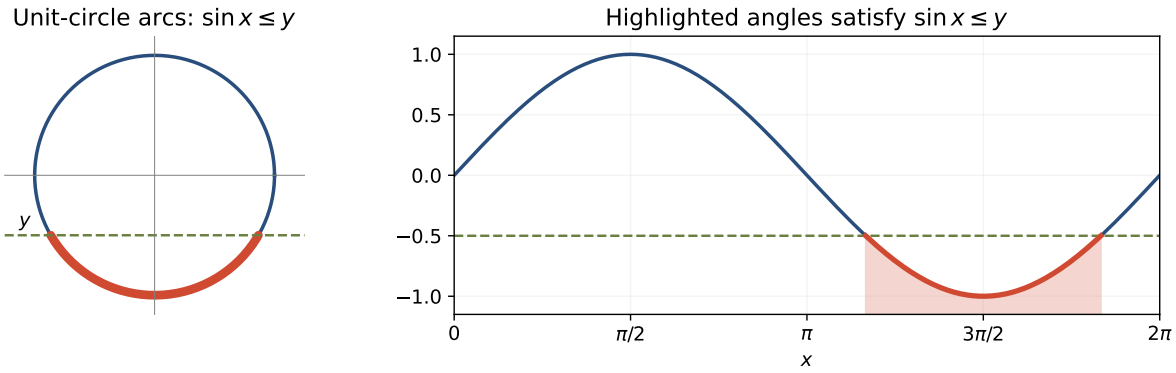


The total highlighted angle measure is $a + [2\pi - (\pi - a)] = \pi + 2a$. Since X is uniform,

$$\begin{aligned} F_Y(y) &= \mathbb{P}(0 < X \le a) + \mathbb{P}(\pi - a \le X < 2\pi) \\ &= \frac{a}{2\pi} + \frac{2\pi - (\pi - a)}{2\pi} = \frac{1}{2} + \frac{\arcsin y}{\pi}. \end{aligned}$$

Case A2: $-1 < y < 0$, so $-\pi/2 < a < 0$. The equality $\sin x = y$ now occurs at $x = \pi - a$ and $x = 2\pi + a$ (*not* at $x = a$, which lies outside the chosen range). The inequality holds for

$$x \in [\pi - a,\ 2\pi + a].$$



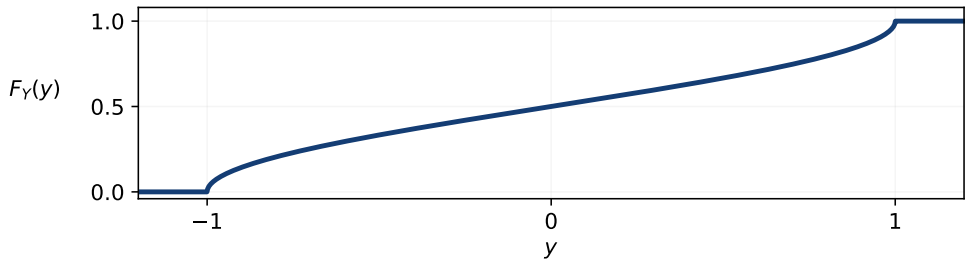
The highlighted interval has length $(2\pi + a) - (\pi - a) = \pi + 2a$. Hence

$$F_Y(y) = \frac{(2\pi + a) - (\pi - a)}{2\pi} = \frac{1}{2} + \frac{\arcsin y}{\pi}.$$

For $y = 0$, the same formula gives $F_Y(0) = 1/2$, as expected from symmetry.

Step 5: Assemble the full CDF of Y.

$$\boxed{F_Y(y) = \begin{cases} 0, & y \le -1, \\ \dfrac{1}{2} + \dfrac{\arcsin y}{\pi}, & -1 < y < 1, \\ 1, & y \ge 1. \end{cases}}$$

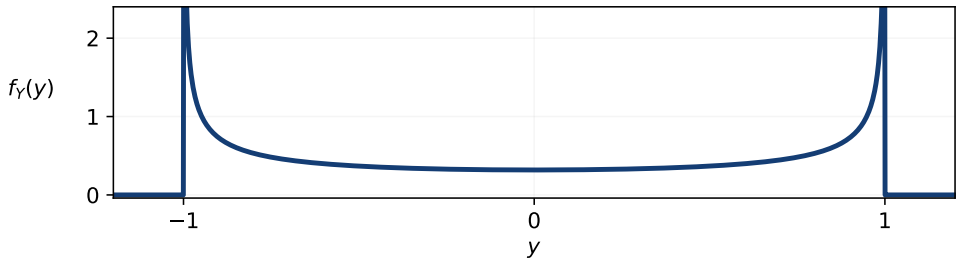


Step 6: Differentiate to obtain the PDF. For $-1 < y < 1$,

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{1}{\pi} \frac{d}{dy} (\arcsin y) = \frac{1}{\pi \sqrt{1 - y^2}}.$$

Therefore

$$\boxed{f_Y(y) = \begin{cases} \dfrac{1}{\pi \sqrt{1 - y^2}}, & -1 < y < 1, \\ 0, & \text{otherwise.} \end{cases}}$$



Note. The density becomes unbounded near $y = \pm 1$, but there are no point masses at either endpoint. Indeed $\int_{-1}^{1} [\pi \sqrt{1 - y^2}]^{-1} \, dy = 1$.